G H PATEL COLLEGE OF ENGINEERING AND TECHNOLOGY

(The Charutar Vidya Mandal (CVM) University)

Department of Information Technology

Program: Computer Science & Engineering (IoT)

VISION

To impart quality education in Information Technology and enable learners to cope with global challenges to build professional career benefitting the sustainable growth of an individual and the society at large.

MISSION

To propose state of the art educational environment equipped with cutting edge technology in the area of Information Technology.

To facilitate learners and faculties with every single opportunity of professional progression embedded in academic scenario itself which can cause enriched workforce contributing to the development of the nation.

To fulfill the noble cause of educating budding technocrats by accelerating the momentum of research and implementing innovative inputs in teaching-learning.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

1. *To build a strong foundation in Computer Science and IoT using modern tools and technologies, while enhancing communication and professional skills.*
2. *To identify real-world problems and develop efficient, cost-effective IoT solutions through systematic analysis, addressing industry and societal needs.*
3. *To grow as ethical professionals with a commitment to teamwork, continuous learning, and adapting to emerging technologies.*

PROGRAMME OUTCOMES (POs)

1. *Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.*
2. *Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.*
3. *Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.*
4. *Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.*
5. *Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with*

an understanding of the limitations.

6. *The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.*
7. *Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.*
8. *Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.*
9. *Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.*
10. *Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.*
11. *Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.*
12. *Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.*

PROGRAMME SPECIFIC OUTCOMES (PSOs)

1. *An ability to identify and analyze real-world problems, and to design IoT-based solutions using appropriate hardware, software, and ethical practices.*
2. *An ability to develop intelligent, connected applications by integrating programming skills, embedded systems, sensor networks, and cloud/edge computing.*
3. *An ability to apply data analytics, cybersecurity principles, and emerging IoT technologies to build secure, scalable, and innovative systems for diverse domains.*

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DEPARTMENT OF INFORMATION TECHNOLOGY

ACADEMIC YEAR: 2026-27; TERM: ODD; SEMESTER: III

DIVISION: CSE-IoT

SUBJECT: Database Management System (102040305)

PRACTICAL INDEX

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Sr. No	Name of the Experiment	Date	Page No.	Signature	Remarks
1	**Creation of Database Objects and Integrity Constraints (DDL). Create database which consist of following table and insert appropriate data into it.** i. Table: regions ii. Table: countries iii. Table: locations iv. Table: warehouses v. Table: employees vi. Table: product_categories vii. Table: products viii. Table: customers ix. Table: orders x. Table: order_items xi. Table: inventories				
2	**Table Cloning and Data Transformation** i. Create warehouse1 from the warehouses table without copying any existing data. ii. Create a full replica of the warehouses table, including all existing records. iii. Create table employee_m from employee using select clause having selected number of columns as employee_id, full_name. full_name column will be merging of first name and last name in columns. Apply constraint to last_column as NOT NULL.				
3	**Schema Evolution and Maintenance** i. Add a salary column to employee_m with an integer data type and a NOT NULL constraint. ii. Modify the salary column to support decimal values using the float(8,2) type. iii. Rename the salary column to salary_e and change its type back to integer. iv. Add a date_of_joining column specifically positioned after the full_name column. v. Perform structural cleanup by dropping the date_of_joining column and renaming the table to employee_r vi. Remove the Primary Key constraint and then truncate all data from the employee_r table.				

4	**Data Manipulation and Basic Retrieval (DML & DQL)** i. Retrieve and display all columns and rows from the products table. ii. Generate a list of unique (distinct) job titles held by employees. iii. Display the city and postalcode for all entries in the locations table.				
5	**Advanced Filtering and Pattern Matching** i. List all employees whose last name contains the substring "ada". ii. Find employees whose last names start with "Jan" or end with the letters "na". iii. Search for employees with a 5-letter last name starting with "D" and having "a" as the third character. iv. Display customer names with a remark "high" if their credit limit is > 2500, else "low". v. Retrieve exactly the first 10 rows and then specifically the 4th row from the products table.				
6	**Sorting, Restricting, and Set Operations (DRL)** i. Display all employee details sorted chronologically by their hiredate. ii. Sort the products table in descending order of listprice and then by category. iii. List countries belonging specifically to Africa, Asia, or America, arranged in that exact regional order. iv. Display name of products belongs to category 3, 1, 5 as per following. v. P1, P2, P3 1 P7, P9 5 P5, P6 vi. List out products with minimum price from each category.				
7	**Scalar and Aggregate Functions** i. Display the current system date with the column label "Date". ii. ii. Calculate a 15% price increase for all products and display it as a whole number labeled "New Price". iii. Display capitalized names and their lengths for employees whose names start with J, A, or M. iv. Calculate the total years of employment for each employee based on their join date. v. Count the total number of employees for each unique job title.				
8	**Complex Joins and Multi-Table Retrieval** i. List every product name alongside its category name, sorted alphabetically by category. ii. Display customer names and the total count of items they have ordered, highest volume first. iii. Find the names of customers who placed orders delivered from the "New Jersey" warehouse. iv. For every order, identify the single most expensive product and list it first. v. List the total number of products sold by each warehouse located in India, ordered				

	alphabetically by warehouse.				
9	**Transactions and Data Security** i. Perform a series of DML updates and use COMMIT to make them permanent. ii. Demonstrate the ROLLBACK command by reverting an accidental data deletion. iii. Create a database View that only shows product names and prices to restrict sensitive cost data. iv. Test the security of a view by attempting to drop it after creation. v. Identify the minimum product price within each category to support low-cost analysis				
10	**PL/SQL Fundamentals and Control Structures** i. Write a basic PL/SQL block to insert a new category into the product_categories table. ii. Create a block using IF-THEN-ELSE to update a product's price based on its current value. iii. Implement a loop that prints the names of the top 5 customers by credit limit.				
11	**Advanced PL/SQL (Cursors, Procedures, and Triggers)** i. Create a Cursor to iterate through and display employees hired in a specific year. ii. Develop a Stored Procedure that accepts a customer_id and returns their total order value. iii. Design a Database Trigger that logs an entry every time a new product is added.				
12	**Mini-Project: System Implementation** i. Design an Entity-Relationship (ER) Diagram based on the provided technology hardware system. ii. Write the full script to create the schema with all required constraints and relationships. iii. Implement at least three complex queries used for monthly business reporting.				

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ASSIGNMENT INDEX

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